

HIPOT TESTERS FOR ELECTRICAL SAFETY

Electrical safety is of utmost importance for all the electrical and electronic products. Today's HIPOT (short for high-potential) testers—key test equipment for electrical safety—have come a long way in terms of features, programmability and safety involved in testing itself



■ SHWETA DHADIWAL BAID

Someone has rightly said that “Safety doesn’t happen by accident.” Today, most consumer electronics devices operate in close vicinity of the human body, be it a mobile phone, laptop, home appliance or medical device. Any product designed without electrical safety parameters in mind may result in a mishap. The finished product needs to be tested thoroughly before it is put in use. Several

tests have to be conducted from design to production stage in order to ensure electrical safety of the component and the finished product.

No designer wants to design an unsafe product, but many a times the product may fail in operation due to some design flaw or faulty components. Electrical safety tests are performed on components such as transformers and switches, cables, motors and finished goods. Several standards organisations publish the permissible

limit for different parameters. Organisations like CSA in Canada, UL/ETL in the United States, GS in Germany, EK in Korea, CCC in China and CNS in Taiwan provide electrical safety/international standards, shares Helena Wang, sales manager, GW Instek. In India, Indian Standards Institute (ISI) specifies electrical safety parameters for many products. Usually, manufacturers hire certified test labs to perform safety tests to comply with these standards and get their products certified.

Insulation plays an important role in electrical safety. During production, most manufacturers include tests around insulation in order to meet safety requirements as per specific standards. As opposed to continuity tests that ensure current flow, there are tests performed using high potential (HIPOT) to ensure that the current does not flow from one point to another at a given voltage for a given time.

Need for electrical safety tests

Electrical safety tests may be mandated by the standards’ organisations, but as a manufacturer, it is your social responsibility. Tests do consume some time in the production line, but these significantly reduce the risk of fires, accidents, electrical shocks, etc from the equipment. It is very important to test insulation as 80 per cent of electrical maintenance and testing involves evaluating insulation integrity.

Electrical insulation starts dete-